

CHUKWUEMEKA OSMUND AZODO

ROTATING EQUIPMENT / MACHINERY RELIABILITY ENGINEER | OIL, GAS & LNG

B.Eng Mechanical Engineering | ASQ Certified Reliability Engineer (CRE) | COREN R.Engr | ASME, SPE, NSE, ASQ
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BOSIET with CA-EBS (valid to 2029) | Offshore Safety Permit (OSP) | Available for immediate international mobilisation

PROFESSIONAL PROFILE

Mechanical engineer with **10+ years' experience** in rotating equipment engineering across offshore gas processing, LNG/NGL trains and gas-fired power generation — currently the **Machinery & Reliability Lead for the Oso Gas Hub**, the largest offshore gas processing facility in the Seplat/JV portfolio. Day-to-day accountability for the **health assessment surveillance of critical rotating equipment** — HP/LP centrifugal gas compressors (5,500+ psi), turbo-expanders, industrial gas turbines, cryogenic and API 610 pumps, diesel engines and lubrication/seal systems — using vibration and lube-oil analytics, performance/compressor maps and digital platforms (SolarInsight, Seeq, PI XHQ). Works shoulder-to-shoulder with operations, maintenance, process, instrumentation and inspection teams to **diagnose machinery problems, optimise intervention scope and duration, and recommend engineered resolutions**. Leads and contributes to **RCFA, risk assessments and technical investigations**; reviews MOCs and project documentation for reliability and operability; and supports **FAT/SAT, commissioning and start-up** of new machinery. Established liaison with OEMs and vendors (Solar Turbines, GE and compressor packagers). Strong working knowledge of **API 610, API 614, API 617, ISO 10816/20816 and ASME PTC-22**. ASQ Certified Reliability Engineer and Category I vibration analyst.

CORE TECHNICAL COMPETENCIES

Rotating Equipment	Centrifugal gas compressors (HP/LP, recompressors) • Turbo-expander/compressor trains • Industrial gas turbines (Solar Taurus/Centaur, GE Frame 9E) • Steam turbines • API 610 centrifugal & cryogenic pumps • Diesel engines / firewater packages • Screw & reciprocating air compressors • Dry gas & mechanical seals, journal/thrust bearings, couplings, lube-oil and seal-gas systems
Health Assessment & Surveillance	Vibration analysis (ISO 10816/20816, Cat-I) • Lube-oil sampling, elemental & particle-count analysis • Infrared thermography • Videoscope/borescope inspection • Performance monitoring via compressor maps, turbine heat rate and pump curves • Deviation flagging, trending and health reporting to stakeholders
Digital Tools & Programs	SolarInsight • Seeq • PI / PI XHQ historian & dashboards • SAP PM (notifications, work orders, strategy) • CMMS/EAM data quality • IIoT sensor-to-cloud (Modbus, OPC-UA) • Python/analytics and AI-assisted failure prediction
Reliability & Problem Solving	Root Cause Failure Analysis (RCFA) • RCA (5-Why, Fishbone, Fault Tree) • FMEA / RCM • Bad-actor and MTBF analysis • Criticality ranking • Life-cycle cost • Reliability improvement and defect-elimination programmes
Projects, MOC & Commissioning	Technical review of vendor/engineering documents, data sheets and deviations • Machinery FAT/SAT witnessing • Pre-commissioning, commissioning and start-up support • PSSR development and execution • Brownfield modification and capacity-expansion support (120 → 240 MMSCFD)
Operations Interface & HSE	Operations/maintenance decision support • Shutdown & turnaround scope optimisation • Risk assessment, HAZID/JSA participation • Permit to Work, LOTO, SIMOPS • OEM and vendor coordination • Multi-discipline technical support and mentoring
Standards	API 610 (centrifugal pumps) • API 614 (lubrication, shaft-sealing and control-oil systems) • API 617 (axial and centrifugal compressors) • ASME PTC-22 (gas turbine performance testing) • ISO 10816 / 20816 (vibration evaluation)

PROFESSIONAL EXPERIENCE

Machinery Engineer Specialist II — Offshore Asset Engineering May 2025 – Present
Am maiden Energy Nigeria Ltd — seconded to **Seplat Energy Producing Nigeria Unlimited** (JV offshore assets: Oso Gas Hub, BRT, UBIT)

Rotating equipment and reliability authority for three offshore assets; deployed February 2026 as dedicated **Machinery/Reliability Lead for the Oso Gas Hub** (120 MMSCFD sales gas + NGL export, expanding to 240 MMSCFD), covering top-tier machinery (HP/LP compressors, recompressors, turbo-expanders, cryogenic pumps), utilities (gas turbine generators, firewater pumps, air compressors) and the NGL fractionation train (de-ethaniser, de-propaniser, de-butaniser).

Equipment Health Surveillance & Technical Support

- Run periodic **health assessment surveillance of critical rotating equipment** — review vibration spectra and trends, lube-oil analysis results, compressor performance maps, turbine parameters and pump deliverability curves — and issue analyses and recommendations to operations, maintenance and asset management for every deviation.
- Exploit digital solutions (**SolarInsight, Seeq, PI XHQ**) to trend machinery condition and process parameters, converting raw data into early-warning calls that prevent unplanned trips and deferment.

- Act as the discipline reference point for other engineering and non-engineering functions — process, instrumentation/controls, inspection, operations, planning, procurement and finance — providing rotating-equipment technical input, spares criticality advice and specification support.
- Provide interim **Asset Engineer coverage**, extending beyond machinery into production optimisation, process stabilisation and non-machinery plant systems — ensuring uninterrupted technical support across the facility.

Machinery Problem Resolution, RCFA & Investigations

- **LP gas compressor (130 MMSCFD, 1,440+ psi) severe vibration after bundle change-out:** led the multi-discipline RCA with OEM and in-house teams — mapped contamination paths, ran sectioned borescope inspections of cooler bundles, identified debris-induced rotor unbalance, then directed process piping cleaning, lube-oil flushing and re-inspection. Machine restored to acceptable ISO vibration limits and returned to service.
- **HP gas compression system (120 MMSCFD, 5,500+ psi):** developed and drove the restoration strategy for the facility's most critical train, safeguarding continuity of gas export to the terminal.
- **30-year lube-oil contamination problem:** a systematic system review exposed as-built vs. as-designed discrepancies behind chronic high differential pressure, rapid filter clogging and element collapse. Introduced an elemental **contamination tracker** with node-by-node sampling to isolate the ingress zones, and coordinated the first comprehensive run-down tank flushing campaign in decades.
- **Depropaniser pump chronic failures (BRT):** RCFA plus an **API 610** review of hydraulic selection, mechanical seal configuration and operating envelope proved the pump was running outside its design point; corrective actions substantially extended run-life and closed out a repeat-failure bad actor.
- **Foundation cracking across multiple pumps and generators (BRT):** led the RCA, established the causal chain and implemented structural reinforcement that restored integrity to the affected rotating equipment foundations.
- **Solar Taurus 60 gas turbine generator failure-to-crank:** supported diagnosis and corrective action with the OEM, restoring power generation availability to the terminal.
- **Firewater diesel engine vibration (UBIT):** resolved a long-standing vibration problem on a safety-critical package and validated performance against pump-curve criteria during deliverability testing.

Risk, Change Review, Projects & Start-up

- Contribute to **risk assessments, HAZID and safety reviews** for machinery interventions, and prepare Lessons Learned under a **PDRR (Prevent, Detect, Respond, Recover)** framework to prevent recurrence.
- Review equipment and system changes (MOC) for their effect on reliability, operability and maintainability; analyse deviations from specification and recommend accept/reject positions.
- Developed and now steward the facility **Pre-Startup Safety Review (PSSR)** process for turbines and compressors — verifying scope completion and safe restart readiness after maintenance and shutdowns.
- Support shutdown and turnaround planning with an emphasis on **optimising machinery scope and duration**, and provide start-up, commissioning and OEM coordination for critical rotating equipment.
- Provide discipline support to the **120 → 240 MMSCFD expansion**, reviewing machinery selection and integration of new equipment into the existing gas processing and NGL trains.

Manager, Business Innovations, Technology & Efficiency

Jul 2022 – May 2025

First Independent Power Limited (FIPL) / Sahara Power Group — multi-site gas turbine power generation

- Led **Reliability-Centred Maintenance (RCM)** deployment across the generating fleet, improving availability and MTBF of critical rotating equipment through revised strategies and defect elimination.
- Supervised **gas turbine performance testing to ASME PTC-22** — power output, heat rate and thermal efficiency verification — and used the results to drive degradation-recovery and washing decisions.
- Managed OEM **Factory Acceptance Tests (FAT)** for critical rotating equipment, witnessing performance and mechanical running tests and closing out punch items before shipment.
- Seconded as **Lead Mechanical/Reliability Engineer for the Imo River Associated Gas Gathering (AGG) plant start-up** — pre-commissioning checks, machinery alignment/verification, first-fire and load-up support.
- Designed and deployed **IIoT-based condition monitoring and predictive maintenance** for gas turbines and compressors, enabling real-time machinery health visibility and delivering ~10% operational cost savings.
- Cut compressor wash downtime by **more than 24 hours** per event through procedure and scope optimisation.

Team Lead, Efficiency & Innovation

Sep 2019 – Jun 2022

First Independent Power Limited (FIPL) / Sahara Power Group

- Led major **gas turbine and compressor overhauls** — planning, clearance verification, rotor and bearing inspection, alignment and re-commissioning — delivering annual savings above NGN 10 million through improved spares management and maintenance planning.
- Chaired cross-functional **RCA investigations** into major turbomachinery failures and drove corrective actions that reduced repeat failures.
- Built real-time plant monitoring dashboards giving operations and maintenance immediate visibility of machinery health, vibration trends and process deviations.
- Coordinated emergency start-ups of critical turbine systems, minimising downtime and restoring generation.

First Independent Power Limited (FIPL) / Sahara Power Group

- Supervised maintenance of high-value turbomachinery and process equipment; delivered quality control during major inspections and overhauls to OEM and industry standards.
- Applied condition monitoring — **vibration analysis, lube-oil analysis, thermography, dissolved gas analysis and motor current signature analysis** — to detect anomalies and prevent unplanned downtime.
- Contributed to review and update of operations and maintenance procedures, strengthening safety protocols and process efficiency.

Graduate Engineer — Mechanical Maintenance

Aug 2014 – Aug 2015

Sahara Power Group — Egbin Power Plc (1,320 MW thermal station)

- Participated in overhauls of steam turbines, gas turbines, diesel engines and balance-of-plant equipment (pumps, compressors, valves, steam traps, accumulators); gained hands-on exposure to demineralisation, water treatment, desalination and lube-oil/water chemistry systems.

Junior Engineer (Intern) — Drillog Petro-Dynamics Ltd

Jun – Dec 2010

- Maintained directional drilling motors and auto-torque equipment and performed QA/QC inspection of mud motors prior to rig deployment — first exposure to oil & gas service operations.

Mechanical Trainee — Industrial Development Centre, Owerri

2008 & 2009

- Machining, bench work, precision alignment of rotating components and fault diagnosis using basic root cause techniques.

EDUCATION**M.Eng Mechanical Engineering (Thermo-Fluids) — in view**

Rivers State University, Port Harcourt

B.Eng Mechanical Engineering — Second Class Upper (2:1), 2012

Federal University of Technology, Owerri (FUTO)

ROTATING EQUIPMENT PORTFOLIO — HANDS-ON EXPERIENCE

Compression	Centrifugal HP gas compressors to 5,500+ psi (120 MMSCFD) • LP compressors 1,440+ psi (130 MMSCFD) • Recompressors • Screw and reciprocating air compressors • Anti-surge control, seal-gas and dry-gas seal systems
Turbines & Drivers	Solar Taurus 60 / Centaur gas turbine generators • GE Frame 9E combined cycle • Steam turbines (Egbin 1,320 MW station) • Diesel engine drivers for firewater and emergency generation • Electric motor drivers and gearboxes
Expanders & Cryogenics	Turbo-expander / booster-compressor trains • Cryogenic pumps • De-ethaniser, de-propaniser and de-butaniser NGL fractionation trains • Sales gas (C1/C2) export and NGL (C3+) export systems
Pumps & Auxiliaries	API 610 centrifugal process pumps • Firewater pump packages and deliverability testing • Lube-oil and seal-oil consoles, run-down tanks, coolers and filtration • Mechanical and dry-gas seals, journal and thrust bearings, couplings

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- ASQ Certified Reliability Engineer (CRE), 2024
- Gas Turbine Performance — Entropy-Tech, 2021
- GE Frame 9E Combined Cycle — SEPCO-Pacific, 2015
- HSE Level I, II & III (OSHA) — HSETrain Intl., 2020
- Basic Rigging & Scaffolding — HSETrain Intl., 2020
- Vibration Analysis Category I — Istec Academy, 2024
- Gas Turbine Mechanical Maintenance — Masaoood John Brown, 2018
- BOSIET with CA-EBS — valid to 2029
- QHSE, Risk & Sustainability Mgmt — IE Safetainability, 2024
- IBM AI Engineering — applied to failure prediction, 2023
- Reliability Engineering Principles — Beina M&R, 2024
- Machinery Vibration Analysis & Alignment — O-Secul, 2019
- Offshore Safety Permit (OSP)
- Basic Instrumentation & Control — Kufman Technical, 2020
- Generative AI & Agents — Johns Hopkins University, 2026

PROFESSIONAL MEMBERSHIPS & RECOGNITION

- American Society for Quality (ASQ), 2024
- COREN Registered Engineer, 2019
- Safety Sentinel Award — FIPL Trans-Amadi, 2023
- Society of Petroleum Engineers (SPE), 2022
- Nigerian Society of Engineers (NSE), 2017
- 1st Runner-Up — Sahara Group Innovation Hackathon, 2020
- American Society of Mechanical Engineers (ASME), 2021
- Safety Advocate of the Year — FIPL, 2023
- Chairman's Recognition for Innovation — Sahara Group, 2019

PERSONAL DETAILS

- Date of birth: 25 September 1988
- Nationality: Nigerian
- Passport: valid — on request

Availability: immediate mobilisation for international assignment. **References:** available on request.